

TOMOKI OKUNO

PhD Candidate in Biostatistics, University of California, Los Angeles (UCLA)

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RESEARCH INTERESTS

- Statistical machine/deep learning for precision and preventive medicine
- Quantitative aging and longevity: DNA methylation biomarkers, causal modeling
- Causal inference & high-dimensional methods

EDUCATION

University of California, Los Angeles

PhD Biostatistics, Department of Biostatistics, UCLA Fielding School of Public Health

Los Angeles, CA

2023 – Expected 2027

- GPA: 4.0/4.0

University of California, Los Angeles

MS Biostatistics, Department of Biostatistics, UCLA Fielding School of Public Health

Los Angeles, CA

2021 – 2023

- GPA: 4.0/4.0
- Master's thesis: "Association of both short-term and long-term glycemic variability with the development of microalbuminuria in the ACCORD trial"

Kyoto University

Master of Informatics, Department of Advanced Mathematical Sciences

Kyoto, Japan

2011 – 2013

- Master's thesis: "Simulation of the Structural Color of Morpho Butterflies with Periodic Fast Multipole Methods"

Kyoto University

Bachelor of Engineering (Computer Science), Faculty of Engineering

Kyoto, Japan

2007 – 2011

- Thesis: "Shape Sensitivity Analysis with FMM in the Periodic Problems for Maxwell's Equation"

AWARDS AND HONORS

- Scholarship for Overseas Graduate Students, Japan Student Services Organization (JASSO), 2023–2027
- Departmental Graduate Fellowship, UCLA, 2023–2026
- Wilshire Fellowship in Geriatric Medicine, UCLA, 2022–2023

PUBLICATIONS

- **Okuno T**, Macwan S, Norman G, Miller D, Reaven P, Zhou J. Continuous Glucose Monitoring Metrics Predict All-cause Mortality in Diabetes: A Real-world Long-term Study. *Diabetes Care* 2025; 48 (10): 1794–1802.
- **Okuno T**, Sort L, Zhang B, et al. Temporal Glycemic Patterns in Type 1 and Type 2 Diabetes: Insights From Extended Continuous Glucose Monitoring. *Journal of Diabetes Science and Technology*. 2025: 19322968251341264.
- **Okuno T**, Macwan SA, Miller D, Norman GJ, Reaven P, Zhou JJ. Assessing patterns of continuous glucose monitoring use and metrics of glycemic control in type 1 diabetes and type 2 diabetes patients in the Veterans Healthcare System:

integrating continuous glucose monitoring device data with electronic health records data. *Diabetes Technol Ther.* 2024; 26(11): 806-813.

- **T Okuno**, A Vansomphone, E Zhang, H Zhou, J Koska, P Reaven, JJ Zhou; Association of both short-term and long-term glycemic variability with the development of microalbuminuria in the ACCORD trial. *Diabetes* 2023; db230374.

RESEARCH EXPERIENCE

Department of Biostatistics at UCLA

Los Angeles, CA

Graduate Student Researcher, Advisor: Dr. Jin J Zhou and Dr. Hua Zhou

March 2022 – Present

- Conducted differential expression analysis with patients' genetic data to predict dysfunction after lung transplantation using Random Forest with time-varying covariates
- Examined the relationship and consistency between data extracted from the electronic health record in the US Veterans Affairs and continuous glucose monitoring data from Dexcom
- Identified real-world, long-term (temporal, seasonal, monthly, and weekly) glycemic trajectories in type 1 and type 2 diabetes from wearable device data
- Examined the association between continuous glucose monitoring device-metrics and all-cause mortality
- Investigated the association of both short-term and long-term glycemic variability with the development of microalbuminuria for type 2 diabetes patients using the Cox model

TEACHING EXPERIENCE

Department of Biostatistics at UCLA

Los Angeles, CA

Teaching Assistant

- BIOSTAT 203B: Introduction to Data Management and Statistical Computing (2022-2024 Winter)
- BIOSTAT 216: Mathematical Methods for Biostatistics (2023-2024 Fall)
- BIOSTAT 401: Data Science Capstone (2025 Spring)

CONSULTING & INDUSTRY EXPERIENCE

Medley Inc. (TSE-Mothers)

Tokyo, Japan

Business Development and Promotion Specialist, Business Enablement Group

2019 – 2021

- Led a 10-member cross-functional team to design and roll out end-to-end processes (sales → contracting → billing → customer support) for a web-based medical questionnaire product; onboarded 200+ clinics in six months
- Built strategic partnerships with medical product manufacturers and distributors, launching cross-sell programs that opened new revenue streams
- Drove inter-departmental operations projects; clarified objectives, KPIs, and risk controls to improve process reliability and transparency
- Conducted market and product due diligence for clinic-facing solutions and translated findings into requirements for commercial and operations teams
- Co-designed and negotiated a public-private telemedicine pilot with the Okinawa prefectural government, delivering services to Yonaguni Island (population ~1,700; one medical facility) and establishing a blueprint for rural deployments

Freelance

Tokyo, Japan

Business Consultant

2018 – 2019

- Collaborated with an e-commerce company and a tax accounting firm
- Managed an e-commerce database of 50,000 stores and led a large-scale system migration; set revised operating costs for the new system
- Built an asset allocation simulator using client profiles; automated data input, analysis, and reporting to cut operating costs and improve sales efficiency

Japan Football Association (JFA)
National Football Team Marketer

Tokyo, Japan
2016 – 2018

- Developed a marketing strategy for JFA; coordinated with ad agencies and JFA sponsors
- Delivered >10% year-over-year fan growth through an engagement program; introduced experiential content to deepen attachment to the National Team
- Built k-means fan segments to surface needs; translated insights into clear targets and campaign objectives
- Planned and executed 2018 FIFA World Cup programming and marketing; secured 10 sponsors and doubled the budget vs the previous World Cup

Accenture Japan Ltd.
Strategy Consultant

Tokyo, Japan
2013 – 2016

- Led 10+ strategy projects across governments and global retailers
- Government: modeled nationwide municipal workloads/costs and secured a 20% increase in baseline compensation
- Government: analyzed Australia/New Zealand programs and drafted a new visa framework to extend high-net-worth visitor stays
- Major retailer: built price/volume forecasting and ROI models; optimized store network and justified doubling the digital promotions budget.

S K I L L S

- **Core Strengths:** Project leadership & ownership, End-to-end program management, Objective decision-making, Precision & high-quality deliverables, Problem solving, willing to rework when evidence demands
- **Methods & Expertise:** Survival and longitudinal analysis, Semiparametric causal inference, High dimensional machine learning; Bayesian hierarchical modeling, Orthogonal and doubly robust estimation
- **Computing Skills:** R, Julia, Python, SAS, Google BigQuery, Quarto, R Shiny, R Markdown, LaTeX